

Highlights

Conformal Surprisal Fusion for Detecting Prompt-Injection Compromise in LLM Agent Traces

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- Fuses observation- and action-side trace signals to detect agent compromise
- Conformal surprisal fusion is scale-free and needs no tuned signal weights
- Attains 0.957 AUROC and 0.919 TPR at a realised 3.4% false-positive rate
- Subtracting the benign maximum provably caps AUROC at one half; it did
- AgentDojo hides no compromise, so its trace-only ceiling is vacuous